

Sreevatsank Kadaveru

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Research Areas: Physical AI | Networked Systems & 5G | Spectrum Sensing

Building Networks and Systems for Physical AI
with Wireless Systems expertise

Education

PhD, *University of California*, San Diego. **2023–Now**
Communication Theory and Systems | PI: Dinesh Bharadia

Master's, *University of California*, San Diego. **2021–2023**
Communication Theory and Systems | ECE Department

Certificate, *Rady School of Management*, San Diego. **2022–2023**
Technology Management Program

Bachelor's, *Indian Institute of Technology*, Indore. **2016–2020**
Electrical Engineering

Experience

WRD, Qualcomm, *Research Intern*, San Diego. **Jun'26–Sept'26**
Developed a rigorous framework to study LLMs for RF+Vision-based multimodal sensing tasks, to identify and isolate failures to gaps in domain knowledge, domain reasoning or domain-specific data interpretation, and to prescribe targeted fine-tuning remedies.

JASR Systems, *Research Intern*, San Diego. **Jun'22–Sept'22**
Blind Detection and Tracking of Low Probability of Intercept Signals.
- Developed a novel statistical approach for wideband spectrum sensing, improving state-of-the-art prediction and enabling multi-receiver capability via an optimal cooperative scheduling algorithm. [Poster]

Tata Digital, *DevOps Engineer*, Mumbai. **Sept'20–Sept'21**
Developed highly automated CI/CD Kubernetes workflows improving reliability and efficiency by 90%.

Télécom Paris, *Summer Research Intern*, France. **May'19–Aug'19**
Improving Reliability of MIMO Mission-Critical Systems using LDPC in 5G.
- Demonstrated a novel latency-reliability trade-off in LDPC code selection for video traffic. [Link]

Research

Systems for Physical AI. **Jun'25–Now**
- **PAIQoO**: Quantifying the network requirements of Physical AI applications
  - Characterising how sensor fidelity and end-to-end latency trade off in robotic task performance.
- **mmsRAN**: Radio Access Networks for Physical AI [Paper]
  - Building an uplink scheduler for multimodal sensor flows that need to be low latency, synchronised and in lock-step order.
- **P2SIF**: Plug-n-Play Sensor Integration Framework [Proposal]
  - Developed a decoupled architecture for new modality integration into existing LLMs without needing to modify LLM weights.

PHY Layer Spectrum Sensing. **2024–2026**
- **PHYNet**: A gold-standard PHY-layer labelled dataset
  - Establishing a standard for evaluating PHY-layer algorithms by building a multi-purpose labelled PHY dataset.
  - Studied the problem of PHY spectrum ignorance in today's world.
- **CamoPHYlage**: Hiding signals in plain sight
  - Developed adversarial mimicking techniques to evade all PHY-layer detectors, enabling surreptitious communication in plain sight.

Spectrum Intelligence Systems, *IARPA-funded project.* **Nov'22–Mar'24**

- **rfYOLO**: Phase modality for better detection
  - Developed a phase transform to improve PHY-layer characterisation of signals using YOLO networks.
- **Nomad**: Architected a high-throughput spectrum-sensing system with a line rate of 40 Gbit/s [Link]
  - Developed containerised microservices with a well-defined, high-throughput shared-memory interface.
  - Achieved major throughput gains with the use of copy-efficient shared memory and a Redis-based in-memory database.

Federated Learning. **Mar'22–Nov'22**

- **Curriculum FL**: Studied the effect of curriculum learning in a federated context. We showed that curriculum learning can significantly improve the convergence time of training. [Paper]

Selected Graduate Projects

Course Projects, *UCSD.* **2021–22**

- **Audio-StyleCLIP**: Extended the capabilities of the StyleCLIP and AudioCLIP models to create a model that can perform style transfer onto an image using an audio styling input. [Report]
- **Convexifying Neural Networks**: Implemented the state-of-the-art technique for modelling a ReLU neural network as a convex optimisation problem, the solution of which is a globally optimal neural network. [Report]
- **Training Robust NN**: Implemented Lipschitz-constant-based regularisation for training robust neural networks and compared performance against L_2 and other baseline techniques. [Report]
- **Lattice Basis Reduction Performance Study**: We evaluated basis reduction algorithms, such as LLL, BKZ, Slide and Self-Dual BKZ, to study the accuracy vs. performance trade-off on COTS hardware. We also studied the scaling of these algorithms with problem size. [Report]

Publications

MobiCom'26: "RoboNEST: Robotics Networked Edge Systems Testbed for Physical AI". [Demo]

MobiCom'26: "WiP: mmsRAN: Multimodal Cellular Uplink for Physical AI". [Paper]

MobiCom'24: "Nomad: Providing Insights into the Spectrum Environment". [Paper]

ICCV'23: "When Do Curricula Work in Federated Learning?" [Paper]

Awards

Qualcomm Innovation Fellowship [Website] *2026*

Powell Foundation Fellowship [Website] *2023*

Shah Family Entrepreneurial Fellowship *2023*

Positions of Responsibility

Teaching Assistant, UCSD *2025–26*

Graduate & Professional Student Association, Representative, UCSD *2023–25*

ECE Graduate Student Council, Board Member, UCSD *2023–25*

Electronics Club, President, IIT Indore *2018–19*

Team Robocon, Technical Lead, IIT Indore *2018–19*

Service

S3 Reviewer & Student Panellist: MobiCom *2026*

Artifact Evaluation Committee: NSDI *2026*

Artifact Evaluation Committee: SIGCOMM *2025*

GPSA Award Committee: UCSD *2024*